

within the range of integration. The game is to obtain the integral as accurately as possible with the smallest number of function evaluations of the integrand. Just as in the case of interpolation (Chapter 3), one has the freedom to choose methods of various *orders*, with higher order sometimes, but not always, giving higher accuracy. *Romberg integration*, which is discussed in §4.3, is a general formalism for making use of integration methods of a variety of different orders, and we recommend it highly.

Apart from the methods of this chapter and of Chapter 17, there are yet other methods for obtaining integrals. One important class is based on function approximation. We discuss explicitly the integration of functions by Chebyshev approximation (*Clenshaw-Curtis quadrature*) in §5.9. Although not explicitly discussed here, you ought to be able to figure out how to do *cubic spline quadrature* using the output of the routine `spline` in §3.3. (Hint: Integrate equation 3.3.3 over x analytically. See [1].)

Some integrals related to Fourier transforms can be calculated using the fast Fourier transform (FFT) algorithm. This is discussed in §13.9. A related problem is the evaluation of integrals with long oscillatory tails. This is discussed at the end of §5.3.

Multidimensional integrals are a whole 'nother multidimensional bag of worms. Section 4.8 is an introductory discussion in this chapter; the important technique of *Monte Carlo integration* is treated in Chapter 7.

CITED REFERENCES AND FURTHER READING:

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- Ralston, A., and Rabinowitz, P. 1978, *A First Course in Numerical Analysis*, 2nd ed.; reprinted 2001 (New York: Dover), Chapter 4.
- Dahlquist, G., and Björck, A. 1974, *Numerical Methods* (Englewood Cliffs, NJ: Prentice-Hall); reprinted 2003 (New York: Dover), §7.4.
- Kahaner, D., Moler, C., and Nash, S. 1989, *Numerical Methods and Software* (Englewood Cliffs, NJ: Prentice-Hall), Chapter 5.
- Forsythe, G.E., Malcolm, M.A., and Moler, C.B. 1977, *Computer Methods for Mathematical Computations* (Englewood Cliffs, NJ: Prentice-Hall), §5.2, p. 89.[1]
- Davis, P., and Rabinowitz, P. 1984, *Methods of Numerical Integration*, 2nd ed. (Orlando, FL: Academic Press).

4.1 Classical Formulas for Equally Spaced Abscissas

Where would any book on numerical analysis be without Mr. Simpson and his “rule”? The classical formulas for integrating a function whose value is known at

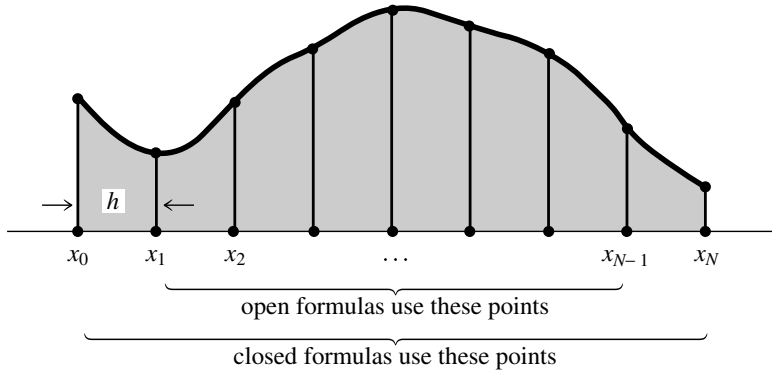


Figure 4.1.1. Quadrature formulas with equally spaced abscissas compute the integral of a function between x_0 and x_N . Closed formulas evaluate the function on the boundary points, while open formulas refrain from doing so (useful if the evaluation algorithm breaks down on the boundary points).

equally spaced steps have a certain elegance about them, and they are redolent with historical association. Through them, the modern numerical analyst communes with the spirits of his or her predecessors back across the centuries, as far as the time of Newton, if not farther. Alas, times *do* change; with the exception of two of the most modest formulas (“extended trapezoidal rule,” equation 4.1.11, and “extended midpoint rule,” equation 4.1.19; see §4.2), the classical formulas are almost entirely useless. They are museum pieces, but beautiful ones; we now enter the museum. (You can skip to §4.2 if you are not touristically inclined.)

Some notation: We have a sequence of abscissas, denoted $x_0, x_1, \dots, x_{N-1}, x_N$, that are spaced apart by a constant step h ,

$$x_i = x_0 + ih \quad i = 0, 1, \dots, N \quad (4.1.1)$$

A function $f(x)$ has known values at the x_i ’s,

$$f(x_i) \equiv f_i \quad (4.1.2)$$

We want to integrate the function $f(x)$ between a lower limit a and an upper limit b , where a and b are each equal to one or the other of the x_i ’s. An integration formula that uses the value of the function at the endpoints, $f(a)$ or $f(b)$, is called a *closed* formula. Occasionally, we want to integrate a function whose value at one or both endpoints is difficult to compute (e.g., the computation of f goes to a limit of zero over zero there, or worse yet has an integrable singularity there). In this case we want an *open* formula, which estimates the integral using only x_i ’s strictly *between* a and b (see Figure 4.1.1).

The basic building blocks of the classical formulas are rules for integrating a function over a small number of intervals. As that number increases, we can find rules that are exact for polynomials of increasingly high order. (Keep in mind that higher order does not always imply higher accuracy in real cases.) A sequence of such closed formulas is now given.

4.1.1 Closed Newton-Cotes Formulas

Trapezoidal rule:

$$\int_{x_0}^{x_1} f(x)dx = h \left[\frac{1}{2} f_0 + \frac{1}{2} f_1 \right] + O(h^3 f'') \quad (4.1.3)$$

Here the error term $O(\)$ signifies that the true answer differs from the estimate by an amount that is the product of some numerical coefficient times h^3 times the value of the function's second derivative somewhere in the interval of integration. The coefficient is knowable, and it can be found in all the standard references on this subject. The point at which the second derivative is to be evaluated is, however, unknowable. If we knew it, we could evaluate the function there and have a higher-order method! Since the product of a knowable and an unknowable is unknowable, we will streamline our formulas and write only $O(\)$, instead of the coefficient.

Equation (4.1.3) is a two-point formula (x_0 and x_1). It is exact for polynomials up to and including degree 1, i.e., $f(x) = x$. One anticipates that there is a three-point formula exact up to polynomials of degree 2. This is true; moreover, by a cancellation of coefficients due to left-right symmetry of the formula, the three-point formula is exact for polynomials up to and including degree 3, i.e., $f(x) = x^3$.

Simpson's rule:

$$\int_{x_0}^{x_2} f(x)dx = h \left[\frac{1}{3} f_0 + \frac{4}{3} f_1 + \frac{1}{3} f_2 \right] + O(h^5 f^{(4)}) \quad (4.1.4)$$

Here $f^{(4)}$ means the fourth derivative of the function f evaluated at an unknown place in the interval. Note also that the formula gives the integral over an interval of size $2h$, so the coefficients add up to 2.

There is no lucky cancellation in the four-point formula, so it is also exact for polynomials up to and including degree 3.

Simpson's $\frac{3}{8}$ rule:

$$\int_{x_0}^{x_3} f(x)dx = h \left[\frac{3}{8} f_0 + \frac{9}{8} f_1 + \frac{9}{8} f_2 + \frac{3}{8} f_3 \right] + O(h^5 f^{(4)}) \quad (4.1.5)$$

The five-point formula again benefits from a cancellation:

Bode's rule:

$$\int_{x_0}^{x_4} f(x)dx = h \left[\frac{14}{45} f_0 + \frac{64}{45} f_1 + \frac{24}{45} f_2 + \frac{64}{45} f_3 + \frac{14}{45} f_4 \right] + O(h^7 f^{(6)}) \quad (4.1.6)$$

This is exact for polynomials up to and including degree 5.

At this point the formulas stop being named after famous personages, so we will not go any further. Consult [1] for additional formulas in the sequence.

4.1.2 Extrapolative Formulas for a Single Interval

We are going to depart from historical practice for a moment. Many texts would give, at this point, a sequence of "Newton-Cotes Formulas of Open Type." Here is

an example:

$$\int_{x_0}^{x_5} f(x)dx = h \left[\frac{55}{24}f_1 + \frac{5}{24}f_2 + \frac{5}{24}f_3 + \frac{55}{24}f_4 \right] + O(h^5 f^{(4)})$$

Notice that the integral from $a = x_0$ to $b = x_5$ is estimated, using only the interior points x_1, x_2, x_3, x_4 . In our opinion, formulas of this type are not useful for the reasons that (i) they cannot usefully be strung together to get “extended” rules, as we are about to do with the closed formulas, and (ii) for all other possible uses they are dominated by the Gaussian integration formulas, which we will introduce in §4.6.

Instead of the Newton-Cotes open formulas, let us set out the formulas for estimating the integral in the single interval from x_0 to x_1 , using values of the function f at $x_1, x_2, \dots$. These will be useful building blocks later for the “extended” open formulas.

$$\int_{x_0}^{x_1} f(x)dx = h[f_1] + O(h^2 f') \quad (4.1.7)$$

$$\int_{x_0}^{x_1} f(x)dx = h \left[\frac{3}{2}f_1 - \frac{1}{2}f_2 \right] + O(h^3 f'') \quad (4.1.8)$$

$$\int_{x_0}^{x_1} f(x)dx = h \left[\frac{23}{12}f_1 - \frac{16}{12}f_2 + \frac{5}{12}f_3 \right] + O(h^4 f^{(3)}) \quad (4.1.9)$$

$$\int_{x_0}^{x_1} f(x)dx = h \left[\frac{55}{24}f_1 - \frac{59}{24}f_2 + \frac{37}{24}f_3 - \frac{9}{24}f_4 \right] + O(h^5 f^{(4)}) \quad (4.1.10)$$

Perhaps a word here would be in order about how formulas like the above can be derived. There are elegant ways, but the most straightforward is to write down the basic form of the formula, replacing the numerical coefficients with unknowns, say p, q, r, s . Without loss of generality take $x_0 = 0$ and $x_1 = 1$, so $h = 1$. Substitute in turn for $f(x)$ (and for f_1, f_2, f_3, f_4) the functions $f(x) = 1$, $f(x) = x$, $f(x) = x^2$, and $f(x) = x^3$. Doing the integral in each case reduces the left-hand side to a number and the right-hand side to a linear equation for the unknowns p, q, r, s . Solving the four equations produced in this way gives the coefficients.

4.1.3 Extended Formulas (Closed)

If we use equation (4.1.3) $N - 1$ times to do the integration in the intervals $(x_0, x_1), (x_1, x_2), \dots, (x_{N-2}, x_{N-1})$ and then add the results, we obtain an “extended” or “composite” formula for the integral from x_0 to x_{N-1} .

Extended trapezoidal rule:

$$\int_{x_0}^{x_{N-1}} f(x)dx = h \left[\frac{1}{2}f_0 + f_1 + f_2 + \dots + f_{N-2} + \frac{1}{2}f_{N-1} \right] + O\left(\frac{(b-a)^3 f''}{N^2}\right) \quad (4.1.11)$$

Here we have written the error estimate in terms of the interval $b - a$ and the number of points N instead of in terms of h . This is clearer, since one is usually holding a and

b fixed and wanting to know, e.g., how much the error will be decreased by taking twice as many steps (in this case, it is by a factor of 4). In subsequent equations we will show *only* the scaling of the error term with the number of steps.

For reasons that will not become clear until §4.2, equation (4.1.11) is in fact the most important equation in this section; it is the basis for most practical quadrature schemes.

The *extended formula of order $1/N^3$* is

$$\int_{x_0}^{x_{N-1}} f(x)dx = h \left[\frac{5}{12}f_0 + \frac{13}{12}f_1 + f_2 + f_3 + \cdots + f_{N-3} + \frac{13}{12}f_{N-2} + \frac{5}{12}f_{N-1} \right] + O\left(\frac{1}{N^3}\right) \quad (4.1.12)$$

(We will see in a moment where this comes from.)

If we apply equation (4.1.4) to successive, nonoverlapping *pairs* of intervals, we get the *extended Simpson's rule*:

$$\int_{x_0}^{x_{N-1}} f(x)dx = h \left[\frac{1}{3}f_0 + \frac{4}{3}f_1 + \frac{2}{3}f_2 + \frac{4}{3}f_3 + \cdots + \frac{2}{3}f_{N-3} + \frac{4}{3}f_{N-2} + \frac{1}{3}f_{N-1} \right] + O\left(\frac{1}{N^4}\right) \quad (4.1.13)$$

Notice that the $2/3$, $4/3$ alternation continues throughout the interior of the evaluation. Many people believe that the wobbling alternation somehow contains deep information about the integral of their function that is not apparent to mortal eyes. In fact, the alternation is an artifact of using the building block (4.1.4). Another extended formula with the same order as Simpson's rule is

$$\int_{x_0}^{x_{N-1}} f(x)dx = h \left[\frac{3}{8}f_0 + \frac{7}{6}f_1 + \frac{23}{24}f_2 + f_3 + f_4 + \cdots + f_{N-5} + f_{N-4} + \frac{23}{24}f_{N-3} + \frac{7}{6}f_{N-2} + \frac{3}{8}f_{N-1} \right] + O\left(\frac{1}{N^4}\right) \quad (4.1.14)$$

This equation is constructed by fitting cubic polynomials through successive groups of four points; we defer details to §19.3, where a similar technique is used in the solution of integral equations. We can, however, tell you where equation (4.1.12) came from. It is Simpson's extended rule, averaged with a modified version of itself in which the first and last steps are done with the trapezoidal rule (4.1.3). The trapezoidal step is *two* orders lower than Simpson's rule; however, its contribution to the integral goes down as an additional power of N (since it is used only twice, not N times). This makes the resulting formula of degree *one* less than Simpson.

4.1.4 Extended Formulas (Open and Semi-Open)

We can construct open and semi-open extended formulas by adding the closed formulas (4.1.11) – (4.1.14), evaluated for the second and subsequent steps, to the

extrapolative open formulas for the first step, (4.1.7) – (4.1.10). As discussed immediately above, it is consistent to use an end step that is of one order lower than the (repeated) interior step. The resulting formulas for an interval open at both ends are as follows.

Equations (4.1.7) and (4.1.11) give

$$\int_{x_0}^{x_{N-1}} f(x)dx = h \left[\frac{3}{2}f_1 + f_2 + f_3 + \cdots + f_{N-3} + \frac{3}{2}f_{N-2} \right] + O\left(\frac{1}{N^2}\right) \quad (4.1.15)$$

Equations (4.1.8) and (4.1.12) give

$$\int_{x_0}^{x_{N-1}} f(x)dx = h \left[\frac{23}{12}f_1 + \frac{7}{12}f_2 + f_3 + f_4 + \cdots + f_{N-4} + \frac{7}{12}f_{N-3} + \frac{23}{12}f_{N-2} \right] + O\left(\frac{1}{N^3}\right) \quad (4.1.16)$$

Equations (4.1.9) and (4.1.13) give

$$\int_{x_0}^{x_{N-1}} f(x)dx = h \left[\frac{27}{12}f_1 + 0 + \frac{13}{12}f_3 + \frac{4}{3}f_4 + \cdots + \frac{4}{3}f_{N-5} + \frac{13}{12}f_{N-4} + 0 + \frac{27}{12}f_{N-2} \right] + O\left(\frac{1}{N^4}\right) \quad (4.1.17)$$

The interior points alternate 4/3 and 2/3. If we want to avoid this alternation, we can combine equations (4.1.9) and (4.1.14), giving

$$\int_{x_0}^{x_{N-1}} f(x)dx = h \left[\frac{55}{24}f_1 - \frac{1}{6}f_2 + \frac{11}{8}f_3 + f_4 + f_5 + f_6 + \cdots + f_{N-6} + f_{N-5} + \frac{11}{8}f_{N-4} - \frac{1}{6}f_{N-3} + \frac{55}{24}f_{N-2} \right] + O\left(\frac{1}{N^4}\right) \quad (4.1.18)$$

We should mention in passing another extended open formula, for use where the limits of integration are located halfway between tabulated abscissas. This one is known as the *extended midpoint rule* and is accurate to the same order as (4.1.15):

$$\int_{x_0}^{x_{N-1}} f(x)dx = h[f_{1/2} + f_{3/2} + f_{5/2} + \cdots + f_{N-5/2} + f_{N-3/2}] + O\left(\frac{1}{N^2}\right) \quad (4.1.19)$$

There are also formulas of higher order for this situation, but we will refrain from giving them.

The *semi-open formulas* are just the obvious combinations of equations (4.1.11) – (4.1.14) with (4.1.15) – (4.1.18), respectively. At the closed end of the integration, use the weights from the former equations; at the open end, use the weights from